Christine P. Lee | Curriculum Vitae, *October 2026*

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Research Interests

My research spans human-computer interaction, artificial intelligence, and robotics, turning people's often-implicit expectations into shared representations that people can shape and AI agents can act on. Accounting for these expectations is important because, as agents increasingly make choices on people's behalf, they can fulfill a request while overlooking priorities or constraints that matter to the person. It is difficult because people often take these expectations for granted, struggling to articulate them in advance and noticing them only when they go unmet, while agents cannot reliably infer what remains unstated. I call the effort of understanding and influencing how agents use these expectations the ***gulf of delegation***, and develop mechanisms that help people specify, check, and revise how their expectations guide agents' choices.

Education

2022– PhD, Computer Sciences, University of Wisconsin–Madison, expected Spring 2027
Advisor: Dr. Bilge Mutlu. *Doctoral Minor:* Communication Arts
Committee: Dr. Bilge Mutlu, Dr. Min Kyung Lee, Dr. Jodi Forlizzi, Dr. Aws Albarghouthi, Dr. Sharon Li

2020–2022 MS, Computer Sciences, University of Wisconsin–Madison
Advisor: Dr. Bilge Mutlu

2015–2019 BS, Computer Sciences, *Summa Cum Laude*, Sejong University, Seoul, South Korea

Research Experience

2020– *Research Assistant*, University of Wisconsin–Madison, Madison, WI
Mentor: Dr. Bilge Mutlu

2026 *Robotics Applied Scientist Co-op*, Fauna Robotics, Amazon, Sunnyvale, CA
Project: Character-based memory architecture for household robots
Mentors: Dr. Hae Won Park, Hani Abdo, Xiaoxu Liu

2020 *Research Intern*, Seoul National University Hospital Biomedical Research Institute, Seoul, South Korea
Mentor: Dr. Taehoon Ko

2019–2020 *Undergraduate Researcher*, CVPR Lab, Sejong University, Seoul, South Korea
Advisor: Dr. Hyun-Joon Moon

Honors & Awards

Fellowships, Awards & Distinctions

2026 Selected Participant, Future Leaders of AI Doctoral Consortium, ACM AI Leadership Summit 2026 (53 selected out of 142)
Position paper: "Designing Responsible AI Workflows: Preserving Control and Supporting Alignment"

Special Recognition for Outstanding Reviews, ACM CHI Conference on Human Factors in Computing Systems (CHI 2026)

2025– NSF Research Traineeship Program "INTEGRATE" Fellowship

2025 Cisco Distinguished Graduate Fellowship, one year of full PhD funding

Special Recognition for Outstanding Reviews, ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)

Special Recognition for Outstanding Reviews, ACM Symposium on User Interface Software and Technology (UIST 2025)

2024 Doctoral Consortium, ACM CHI Conference on Human Factors in Computing Systems (CHI 2024), with $1,700 award

2021 Summer Research Assistant Award, UW–Madison Computer Sciences Department
Chosen as one of 15 students in the department based on faculty nomination

Paper Awards

2025 Honorable Mention (Top 251 in 5020), ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)
Kim, Sato, White, Ho, **Lee**, Hwang, & Mutlu: Bridging Generations using AI-Supported Co-Creative Activities

2022 Honorable Mention (Top 131 in 2597), ACM CHI Conference on Human Factors in Computing Systems (CHI 2022)
Lee, Cagiltay, & Mutlu: The Unboxing Experience: Exploration and Design of Initial Interactions Between Children and Social Robots

Publications

* indicates equal contribution. 🏆 indicates a paper award (see Paper Awards).

Full-Length Conference Papers

Lee, C. P., Porfirio, D., Pan, E., & Mutlu, B. PromptPath: Bringing Persistent User Context into LLM Prompt Formulation. *Under review*.

Lee, C. P., Sullivan, D., Wang, X. J., & Mutlu, B. A Communication Initialization Profile (CIP) for Cold-Start LLM Personalization. *Under review*.

Xu, M. F., **Lee, C. P.**, He, X., & Mutlu, B. Responding to Challenging Inputs: Comparing Strategies for Public LLM-Powered Robots. *Under review*.

Xu, M. F., He, X., **Lee, C. P.**, & Mutlu, B. Speaking To, Around, and About: Public Encounters with LLM-Powered Conversational Robots. *Under review*.

2026 **Lee, C. P.**, Lee, M. K.*, & Mutlu, B.* From Tool to Agent: How Worker Needs Reorganize Across the Agentic Roles of Workplace AI. To appear in *Proceedings of the 2026 International Conference on Human-Agent Interaction (HAI '26)*.

Wang, X. J., **Lee, C. P.**, & Mutlu, B. LearnMate^2: Design and Evaluation of an LLM-powered Personalized and Adaptive Support System for Online Learning. *Proceedings of the 2026 ACM Designing Interactive Systems Conference (DIS '26)*.

Kim, Y., **Lee, C. P.**, & Mutlu, B. Speaking with Screens: Design Space and Guidelines for Informational Robot Screens. *Proceedings of the 2026 ACM/IEEE International Conference on Human-Robot Interaction (HRI '26)*.

2025 **Lee, C. P.**, Porfirio, D., Wang, X. J., Zhao, K., & Mutlu, B. VeriPlan: Integrating Formal Verification and LLMs into End-User Planning. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*.

🏆 Kim, C. Y., Sato, A. J., White, N. T., Ho, H.-R., **Lee, C. P.**, Hwang, Y., & Mutlu, B. Bridging Generations using AI-Supported Co-Creative Activities. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*.

2024 **Lee, C. P.**, Lee, M. K.*, & Mutlu, B.* The AI-DEC: A Card-based Design Method for User-centered AI Explanations. *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)*.

Lee, C. P., Praveena, P., & Mutlu, B. REX: Designing User-centered Repair and Explanations to Address Robot Failures. *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)*.

Kim, C. Y.*, **Lee, C. P.***, & Mutlu, B. Understanding Large-Language Model (LLM)-powered Human-Robot Interaction. *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI '24)*.

2022 **Lee, C. P.**, Cagiltay, B., & Mutlu, B. The Unboxing Experience: Exploration and Design of Initial Interactions Between Children and Social Robots. *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*.

Journal Articles

2026 Hwang, Y., **Lee, C. P.**, Porfirio, D., Hiatt, L. M., & Mutlu, B. Formal Methods in Robot End-User Development: Progress, Gaps, and Opportunities. *Foundations and Trends in Robotics*, 13(3), 273–349.

2020 Kim, H. K., Choi, S. W., Bae, Y. S., Choi, J., Kwon, H., **Lee, C. P.**, Lee, H.-Y., & Ko, T. MARIE: A Context-Aware Term Mapping with String Matching and Embedding Vectors. *Applied Sciences*, 10(21), 7831.

Short Papers & Extended Abstracts

Lee, C. P., Liu, X., Abdo, H., & Park, H. W. Character-Based Memory: Shaping How Household Robots Interpret Experience. *Under review*.

2025 **Lee, C. P.***, Choi, J.*, & Mutlu, B. MAP: Multi-user Personalization with Collaborative LLM-powered Agents. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*.

Wang, X. J., **Lee, C. P.**, & Mutlu, B. LearnMate: Enhancing Online Education with LLM-Powered Personalized Learning Plans and Support. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*.

2024 **Lee, C. P.** Design, Development, and Deployment of Context-adaptive AI Systems for Enhanced User Adoption. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*.

2023 **Lee, C. P.**, Cagiltay, B., Sullivan, D., & Mutlu, B. Demonstrating the Potential of Interactive Product Packaging for Enriching Human-Robot Interaction. *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI '23)*.

Preprints

2026 **Lee, C. P.**, Wang, X. J., Albarghouthi, A., Porfirio, D., & Mutlu, B. U-Define: Designing User Workflows for Hard and Soft Constraints in LLM-Based Planning. *arXiv preprint*, arXiv:2605.02765.

Other Publications

2019 Nam, J., **Lee, C. P.**, Patankar, A. A., Wang, H., Li, Y., & Moon, H. Object Classification for Domestic Waste based on Convolutional Neural Networks. *Proceedings of the Korean Institute of Broadcast and Media Engineers Fall Conference*.

2018 Jung, J. Y., **Lee, C. P.**, Kim, J. H., Kim, S. I., Kim, J. I., Kong, H. S., Lee, Y. B., Lee, Y. L., & Kang, J. W. Increasing Video Compression Efficiency using Residual Data Distribution and Transform. *Proceedings of the WISET Junior Science & Technology Research Reports*, Vol. 1, pp. 350–354.

Talks

Invited Talks

2025 *Ploutos AI Community*, April 2025
"VeriPlan: Integrating Formal Verification and LLMs into End-User Planning"

Carnegie Mellon University, HRI Reading Group, March 2025
"Formal Methods for User-Centered AI: Enabling Effective Real-World Integration"

2024 *The University of Texas at Austin*, School of Information, hosted by Dr. Min Kyung Lee, September 2024
"Designing User-Centered Tools for Real-World AI Integration"

Conference Presentations

2025 *ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)*, Yokohama, Japan
"VeriPlan: Integrating Formal Verification and LLMs into End-User Planning"

2024 *ACM Designing Interactive Systems Conference (DIS 2024)*, Copenhagen, Denmark
"The AI-DEC: A Card-based Design Method for User-centered AI Explanations"

ACM Designing Interactive Systems Conference (DIS 2024), Copenhagen, Denmark
"REX: Designing User-centered Repair and Explanations to Address Robot Failures"

2022 *ACM CHI Conference on Human Factors in Computing Systems (CHI 2022)*, New Orleans, LA
"The Unboxing Experience: Exploration and Design of Initial Interactions Between Children and Social Robots"

Teaching & Mentoring

Teaching

2026 *Guest Instructor (two sessions)*, CS 639: Interaction Design Studio, UW–Madison, March 2026
"Proactivity & Timing" and "Collaboration & GenAI"

2025 *Guest Lecture*, Autonomous Robotics, UW–Madison, April 2025
"Introduction to Human-Robot Interaction (HRI)"

2024 *Guest Lecture (two sessions)*, Graduate Human-AI Interaction course, The University of Texas at Austin, Fall 2024
Hands-on session with the AI-DEC design method

2020–2021 *Teaching Assistant*, CS 354: Machine Organization and Programming, UW–Madison, Fall 2020 and Spring 2021
Instructor: Dr. Debra Deppeler

Student Mentoring

2025– *Ethan Pan*, UW–Madison
Undergraduate research on system development for persistent user context in LLM prompting

2024– *Xinyu Jessica Wang*, UW–Madison (now PhD student, UW–Madison)
Undergraduate honors thesis on LLM-powered personalized learning support, leading to LearnMate^2 (DIS '26)

2024–2025 *Kevin Zhao*, UW–Madison (now Software Development Engineer, Amazon Web Services)
Undergraduate honors thesis on LLM-assisted model checking for end-user planning

2022–2023 *Hongyu Fu*, UW–Madison (now PhD student, Carnegie Mellon University)
Undergraduate research on route planning for recovery from robot failures

Colin Li, UW–Madison (now PhD student, University of Southern California)
Undergraduate research on LLM-based robot navigation

Outreach

2024 UW–Madison Grandparents University
Helper for a two-day workshop in which grandparents and grandchildren co-created stories with AI and programmed a social robot (three weekly cohorts, 67 participants)

2023 High school outreach
Hosted half-day lab visit for robot demonstration and discussion panel for computer science majors

Service

Community

2025–2026 Webmaster, Human-Computer Interaction Consortium (HCIC), https://hcic.org

Department

2024–2025 Graduate Admissions Application Reader, PhD Program, Department of Computer Sciences, UW–Madison
Fall 2024 and Fall 2025 admissions cycles, about 200 applications per cycle

Conference Reviewing

2022– ACM CHI Conference on Human Factors in Computing Systems (CHI)

2023– ACM/IEEE International Conference on Human-Robot Interaction (HRI)

2023– ACM Designing Interactive Systems Conference (DIS)

2024– ACM Symposium on User Interface Software and Technology (UIST)

2026– ACM International Conference on Human-Agent Interaction (HAI)

Journal Reviewing

ACM Transactions on Computer-Human Interaction (TOCHI)

ACM Transactions on Human-Robot Interaction (THRI)

Workshop & Symposium Reviewing

2025 IEEE International Conference on Robotics and Automation (ICRA 2025), Workshop on "Human-Centered Robot Learning in the Era of Big Data and Large Models"

2024 AAAI Fall Symposium Series (FSS 2024), Symposium on "Unifying Representations for Robot Application Development" (UR-RAD)

Selected Press

2024 *AIhub*, Bridging the Gap between User Expectations and AI Capabilities: Introducing the AI-DEC Design Tool, August 2024

References

Bilge Mutlu, Stephen Wright Professor, Department of Computer Sciences, University of Wisconsin–Madison
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Min Kyung Lee, Associate Professor, School of Information, The University of Texas at Austin
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Jodi Forlizzi, Herbert A. Simon Professor, Human-Computer Interaction Institute, Carnegie Mellon University
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Hae Won Park, Amazon Scholar, Amazon
robohwp@amazon.com